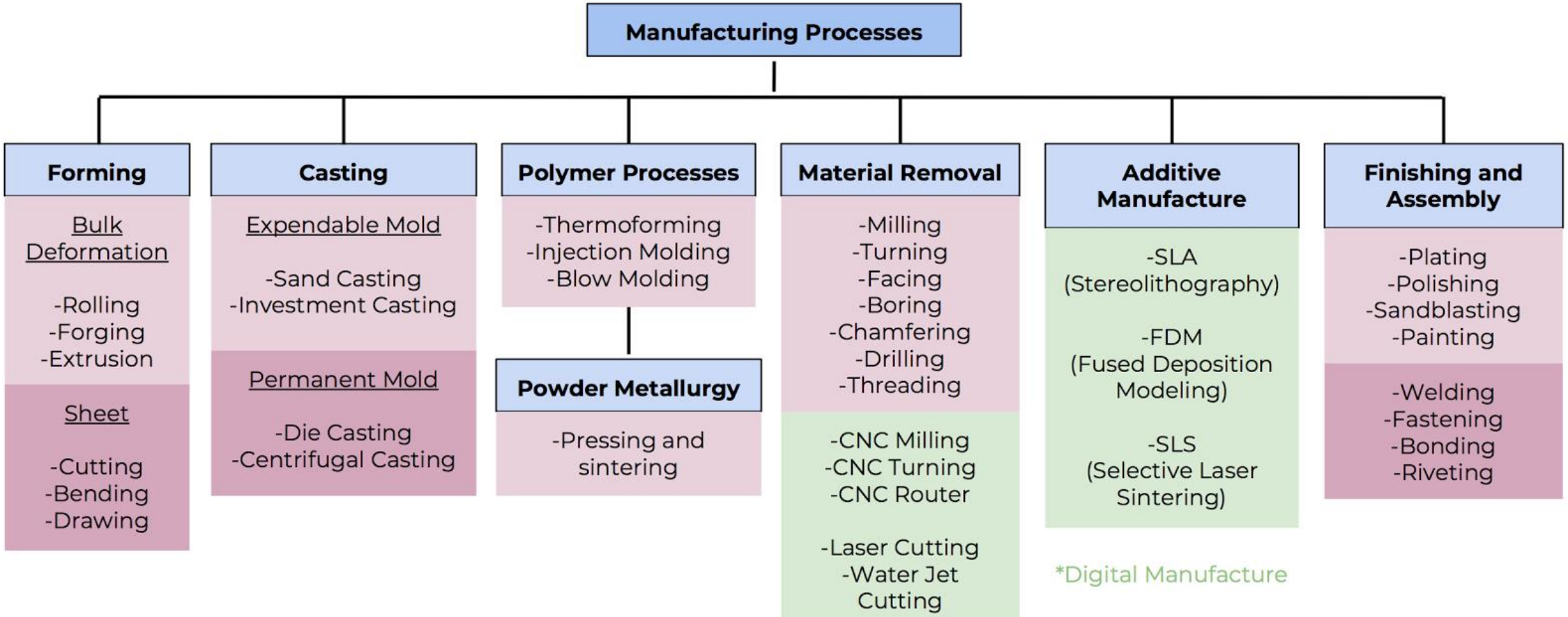
# Experiment No:4

Hardness Measurement of Arc Melted Metallic Alloy


Manufacturing Processes
Forming
Bulk Deformation
-Rolling
-Forging
-Extrusion
Sheet
-Cutting
-Bending
-Drawing
Casting
Expendable Mold
-Sand Casting
-Investment Casting
Permanent Mold
-Die Casting
-Centrifugal Casting
Polymer Processes
-Thermoforming
-Injection Molding
-Blow Molding
Powder Metallurgy
-Pressing and sintering
Material Removal
-Milling
-Turning
-Facing
-Boring
-Chamfering
-Drilling
-Threading
-CNC Milling
-CNC Turning
-CNC Router
-Laser Cutting
-Water Jet Cutting
Additive Manufacture
-SLA (Stereolithography)
-FDM (Fused Deposition Modeling)
-SLS (Selective Laser Sintering)
*Digital Manufacture
Finishing and Assembly
-Plating
-Polishing
-Sandblasting
-Painting
-Welding
-Fastening
-Bonding
-Riveting

# Melting & Pouring Temperatures

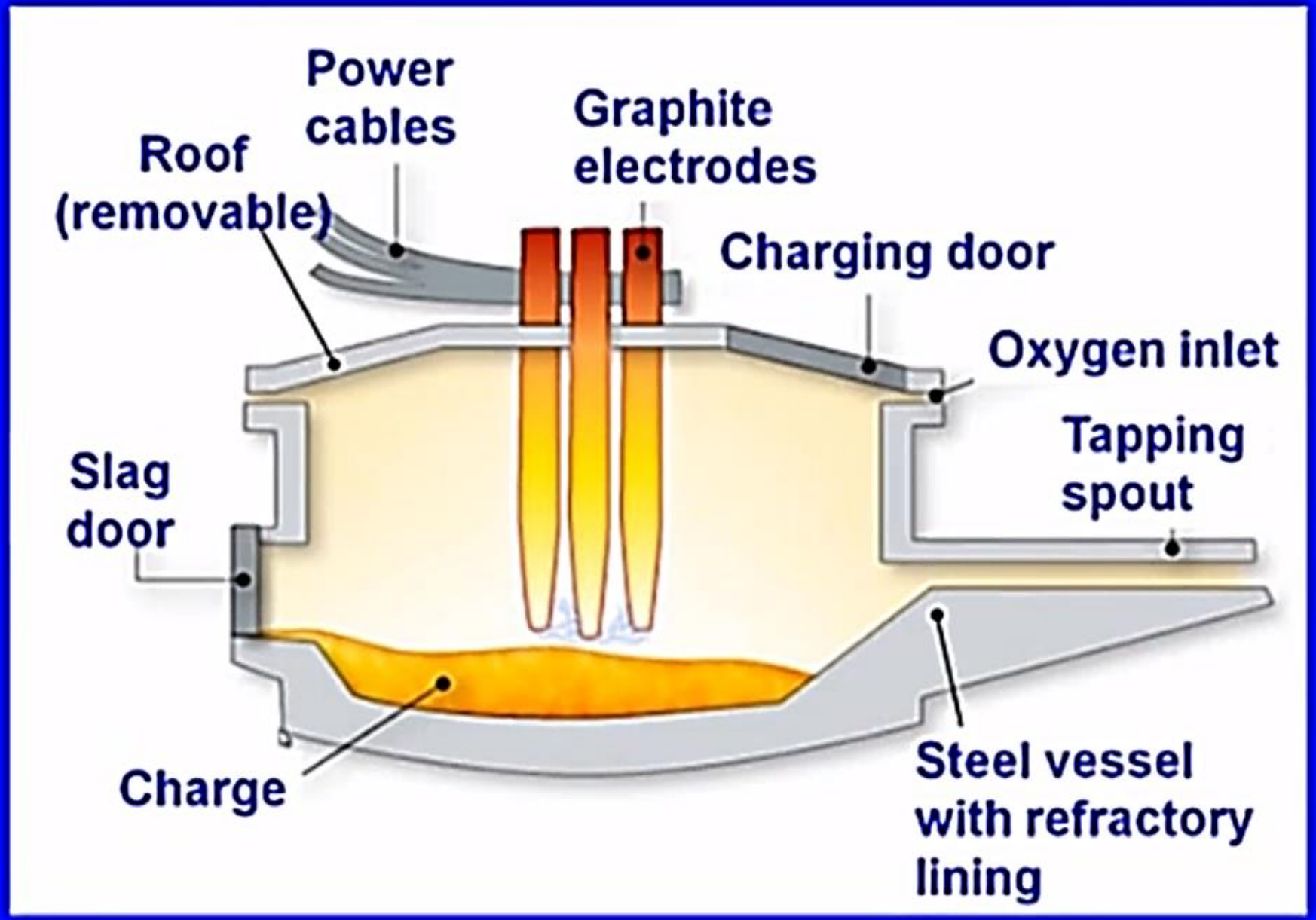
| S. No | Metal / Alloy | Melting temperature (°C) | Pouring temperature range (°C) |
|---|---|---|---|
| 1. | Gray Cast Iron | 1370 | 1510 – 1590 |
| 2. | Cast steel | 1480 | 1600 – 1720 |
| 3. | Copper | 1083 | 1130 – 1200 |
| 4. | Nickel | 1453 | 1500 – 1590 |
| 5. | Aluminum | 660 | 700 – 760 |
| 6. | Zinc | 420 | 450 – 480 |
| 7. | Lead | 327 | 350 – 380 |
| 8. | Tin | 232 | 280 – 290 |
| 9. | Cu- 4.0 Ni alloy | 1175 | 1220 – 1280 |
| 10. | Gun metal (Cu-85%, Sn-5%, Zn-5%, Pb-5%) | 1040 | 1100 – 1180 |

# Important furnaces used in melting

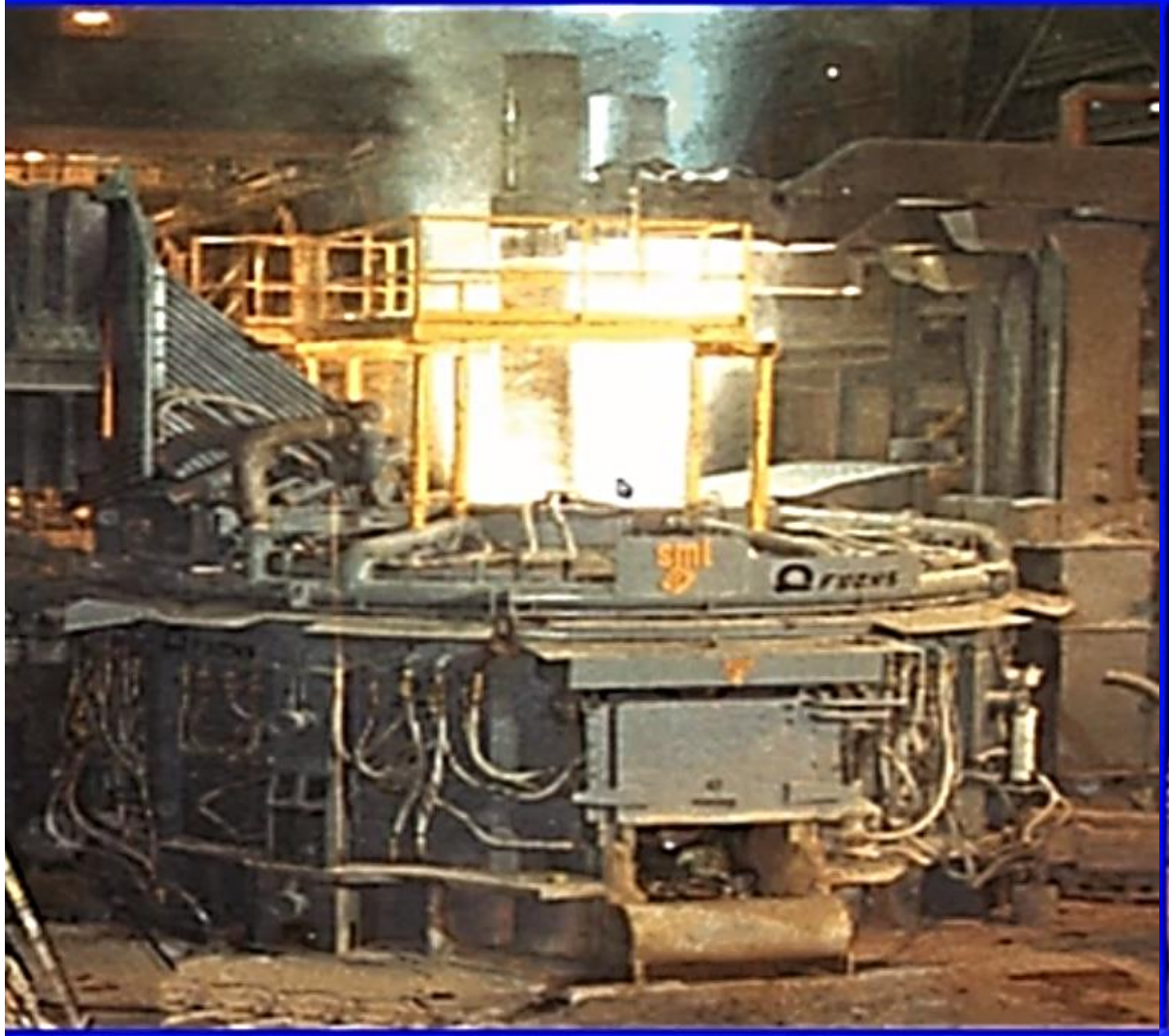
1. Crucible furnace
2. Cupola furnace
3. Electric arc furnace
4. Induction furnace
5. Resistance furnace
6. Rotary furnace
7. Reverberatory furnace

| | FURNACE | MODE OF MELTING | APPLICATION |
|---|---|---|---|
| 1. | Crucible furnace | Solid fuel, Oil or Gas | Most of alloys except steel |
| 2. | Cupola furnace | Coke, Oil | Cast iron, Steel |
| 3. | Electric arc furnace | | |
| | Single phase | Arc on metal charge | Non-ferrous alloys |
| | Three phase | Arc on metal charge | Steels |
| 4. | Induction furnace | | |
| | Low frequency | Electromagnetic induction | Non-ferrous alloys |
| | High frequency | Electromagnetic induction | Steel and alloy steels |
| 5. | Resistance furnace | Resistance caused to the current | Steel, Non-ferrous alloys |
| 6. | Rotary furnace | Pulverized solid fuel, Gas or Oil | Non-ferrous alloys, Cast iron |
| 7. | Reverberatory furnace | Solid fuel, Gas or Oil | Non-ferrous alloys, Cast iron |

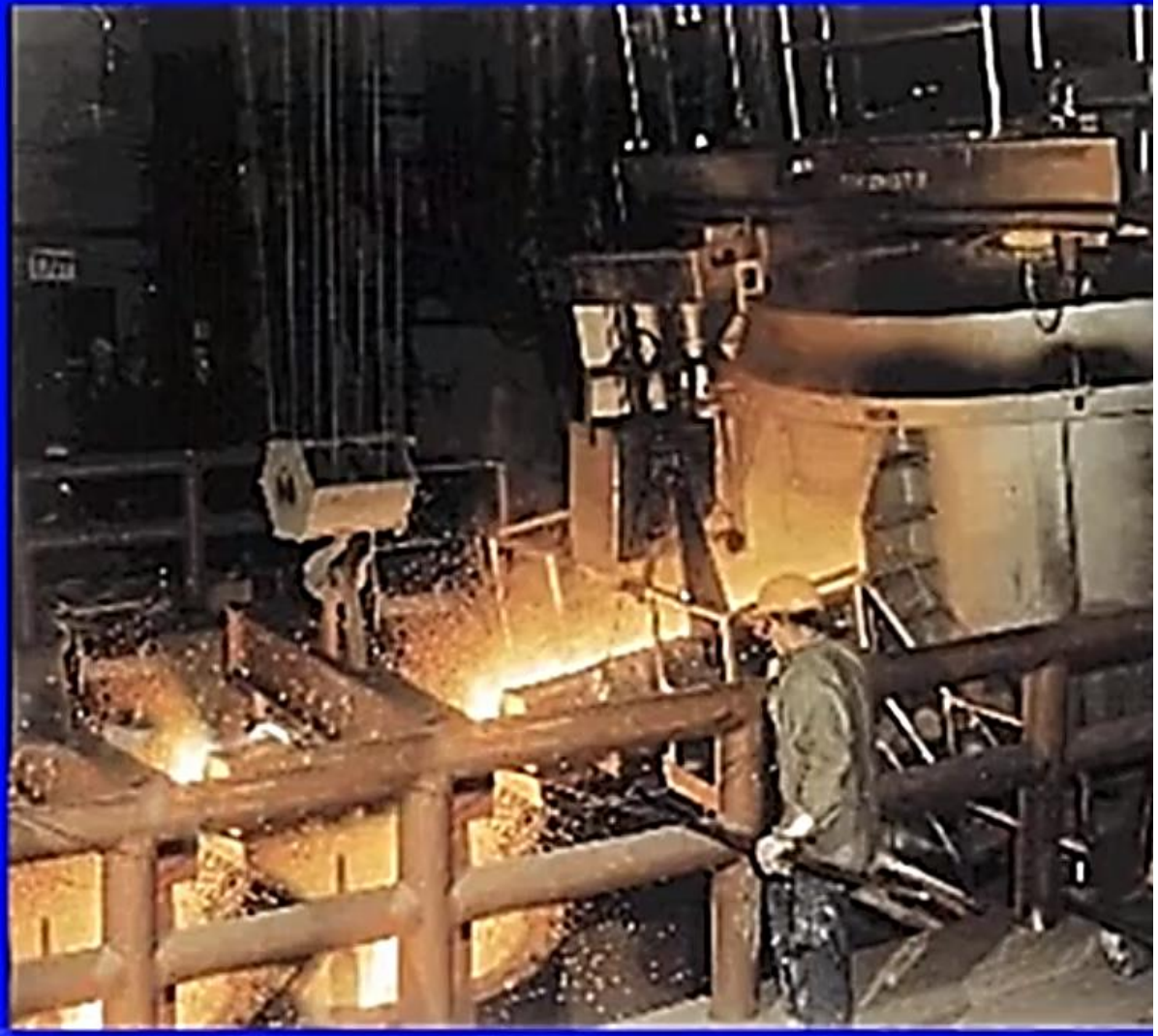
# ELECTRIC ARC FURNACE



## Arc is intensified



## Tapping of molten metal



# Vacuum Arc Melting Unit

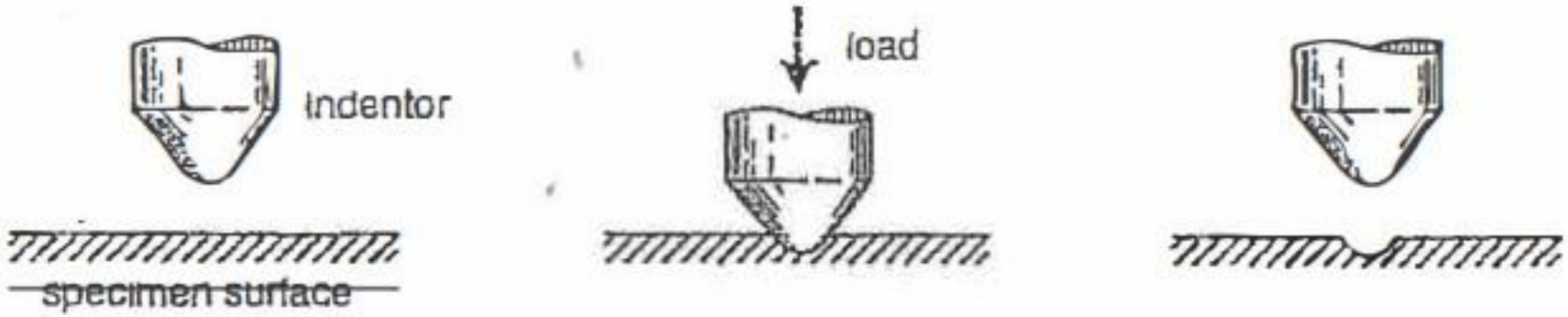
- Arc Melting is used for melting metals– typically to form alloys. Heating is via an electric arc struck between a tungsten electrode and metals placed in a depression (crucible) in the copper hearth.
- In vacuum arc melting the chamber is evacuated and then back filled with argon gas. Hence, melting is performed in argon atmosphere.
- Basic Principle: A standard Tungsten Inert Gas (TIG) welding unit is used as a power source. Heat generated by the electric arc struck between the electrode and the metals serves to melt the metals placed in the crucible to form an alloy. Repeated melting is performed to improve the homogeneity of the alloy. Evacuation of the chamber avoids oxidation of the melt (Ar being an inert gas does not react with molten metal).
- The metals can be heated to a temperature in excess of 2000ºC.

ARC MELTING FURNACE
SCIENTIFIC & ANALYTICAL INSTRUMENTS
NEW DELHI
WATER CHILLER

# Arc melting demonstration video
# (Please watch on BB)

# Arc melting demonstration video
# (Please watch on BB)

# Hardness Measurement



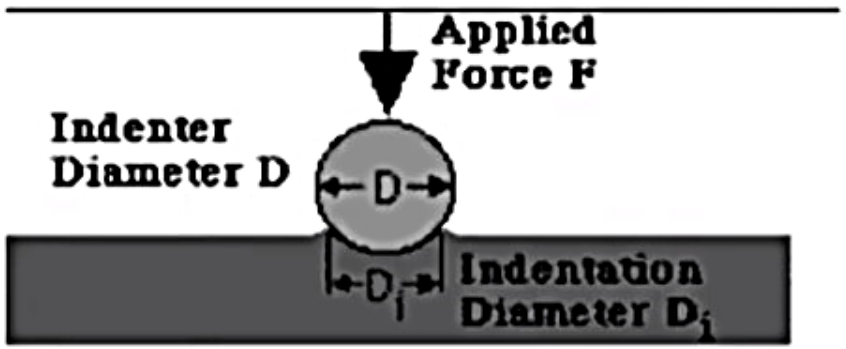
# Brinell Hardness

- Indentation is done with 10 mm diameter steel ball.
- A load of 3000 kg (500 kg for softer materials) is applied for 10 – 30 s.
- Dia of the indentation is measured to obtain the hardness (Brinell Hardness No.) from the relationship

Brinell Hardness No. (BHN)

$$BHN = \frac{P}{(\pi D/2)(D - \sqrt{D^2 - d^2})} = \frac{P}{\pi D t} \qquad (kg/m^2)$$

P = Applied load, Kg
D = Diameter of ball, mm
d = Dia of indentation, mm
t = Depth of impression, mm

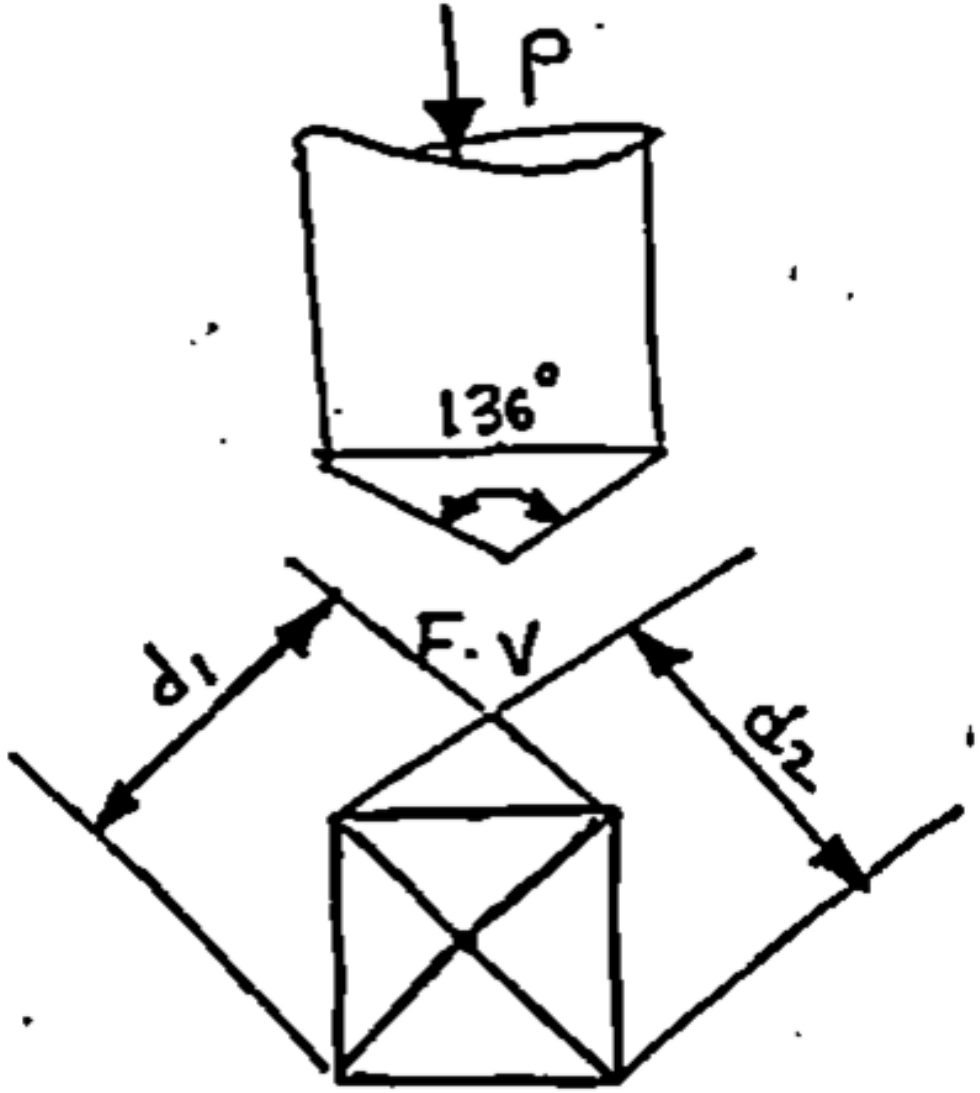


# Vickers hardness

- Uses a square-base diamond pyramid as indenter having angle as 136° between opposite faces of the pyramid
- Vickers hardness number (VHN) is the load divided by the surface area of the indenter
- Area measured from microscopic measurements of the lengths of the diagonals of the indentation

$$\mathbf{HV} = \frac{2P\sin(\theta/2)}{L^2} = \frac{1.854P}{L^2}$$

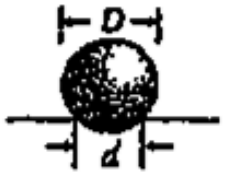
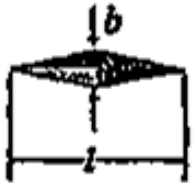
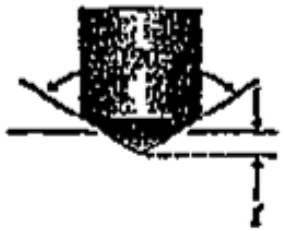
$P$ = applied load, Kg
$L$ = average length of diagonals, mm
θ = angle between opposite faces of diagonal = 136°


P
136°
d1
F.V
d2

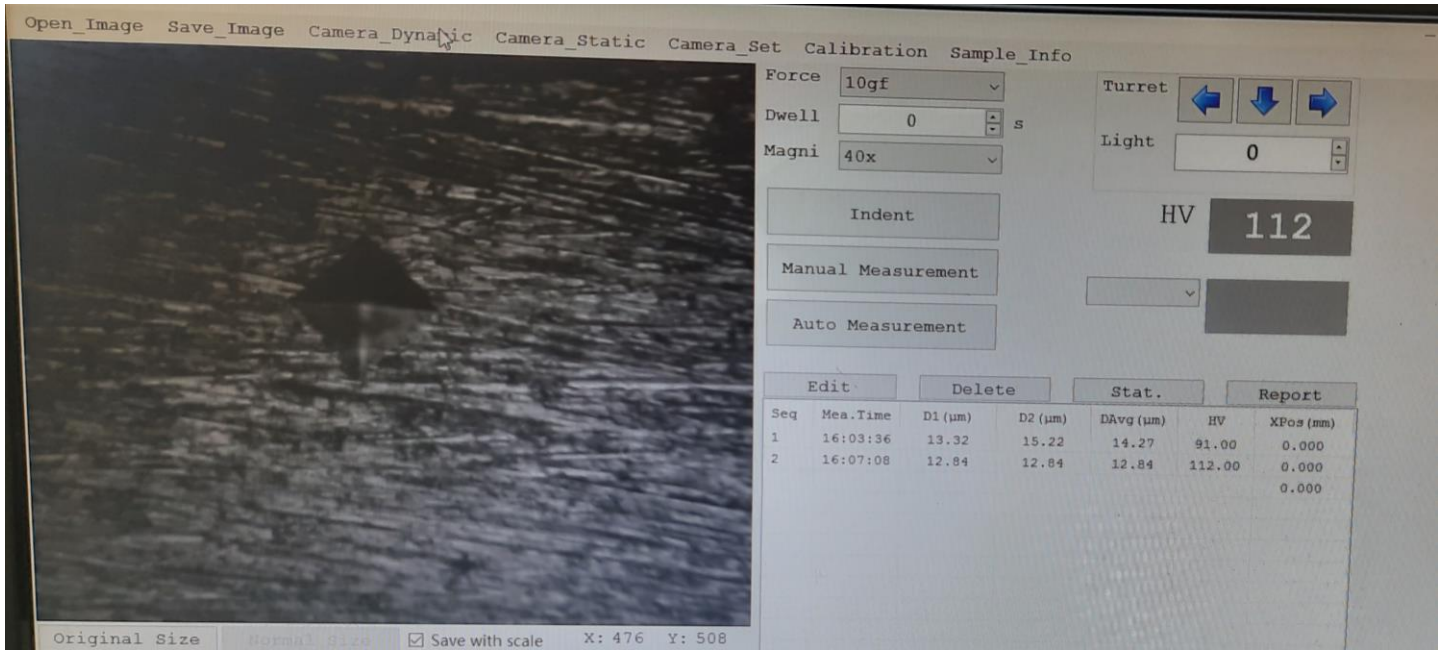
# Rockwell hardness

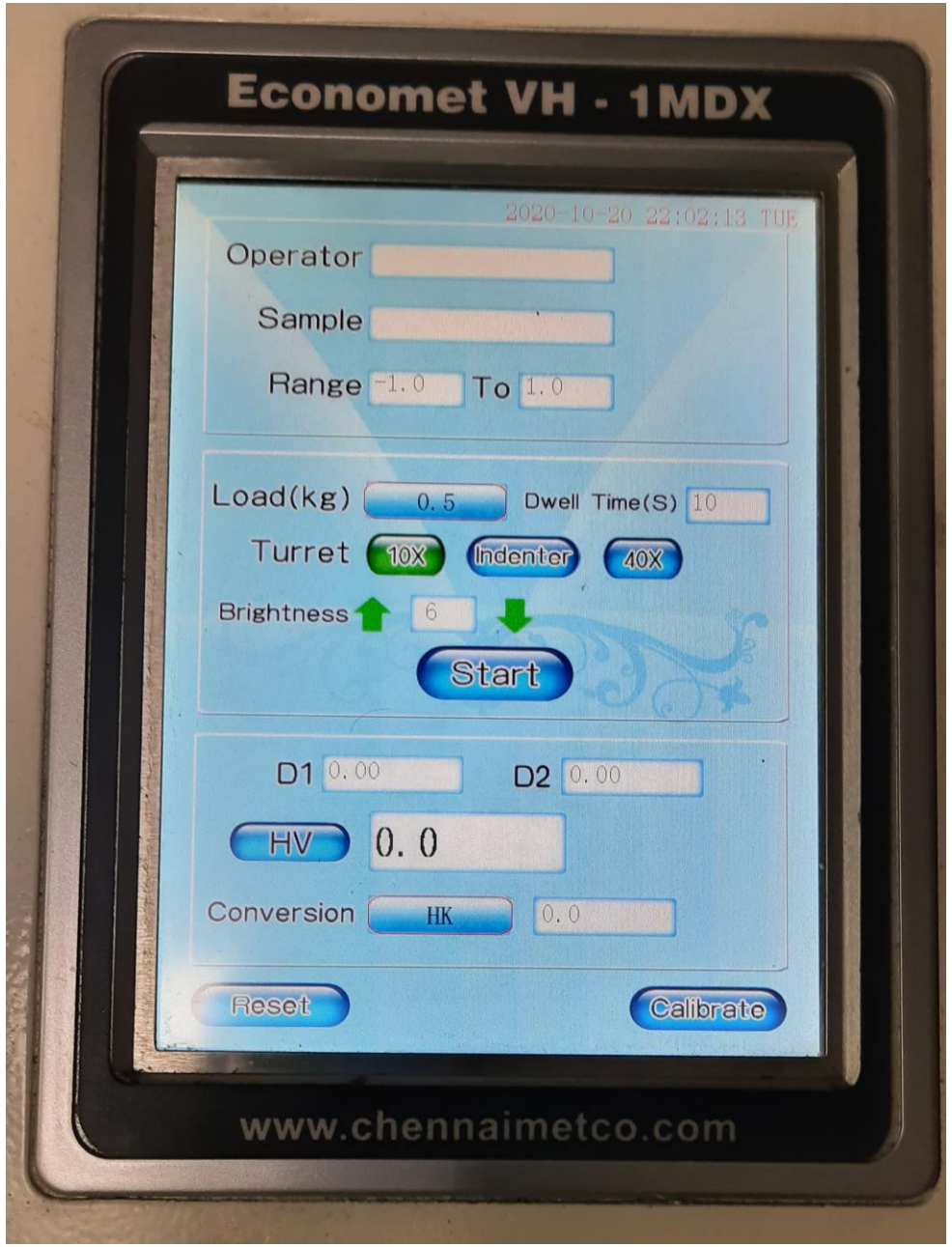
- In this type of test, depth of indentation at a constant load is taken as the measure of Hardness.
- A minor load of 10 kg is first applied for good contact between the indenter and the sample surface.
- The major load is then applied and the depth of indentation is recorded on a dial gage in terms of an arbitrary number.
- The dial consists of 100 divisions, each division representing a penetration depth of 0.002 mm.

shape of indentation :

| test | indenter | side view | top view |
|---|---|---|---|
| Brinell | 10 mm sphere |   | |
| Vickers | diamond pyramid | | |
| Knoop microhardness | diamond pyramid | |   |
| Rockwell | | | |
| A<br>C | diamond cone |   | |
| B | 1.6 mm diameter sphere | | |

# Vickers' Hardness demonstration video (Please watch on BB)


Open_Image
Save_Image
Camera_Dynamic
Camera_Static
Camera_Set
Calibration
Sample_Info
Force
10gf
Dwell
0
s
Magni
40x
Turret
Light
0
Indent
HV
112
Manual Measurement
Auto Measurement
Edit
Delete
Stat.
Report
Seq Mea.Time D1 (µm) D2 (µm) DAvg (µm) HV XPos (mm)
1 16:03:36 13.32 15.22 14.27 91.00 0.000
2 16:07:08 12.84 12.84 12.84 112.00 0.000
0.000
Original Size
Save with scale
X: 476 Y: 508


Economet VH - 1MDX
2020-10-20 22:02:13 TUE
Operator
Sample
Range -1.0 To 1.0
Load(kg) 0.5 Dwell Time(S) 10
Turret 10X Indenter 40X
Brightness 6
Start
D1 0.00 D2 0.00
HV 0.0
Conversion HK 0.0
Reset
Calibrate
www.chennaimetco.com